that balance seems like a good idea. Unfortunately, measuring the matter content of the universe is complicated by the fact that not all matter is easy to see, and determining how much a galaxy weighs when all you have is a picture of it can be challenging at best. As far back as the 1930s, it was clear that just counting up galaxies and stars meant missing something important. Astronomer Fritz Zwicky studied the motions of galaxies moving around in galaxy clusters and noticed they seemed to be moving too quickly, and should by rights be flying off into empty space, like children on a merry-go-round that's spinning too fast. He suggested that perhaps there was some unseen "dark matter" holding everything together. That idea floated around the astronomical community as an unsettling possibility until sometime in the 1970s, when Vera Rubin came along and demonstrated once and for all that whole heaps of spiral galaxies really didn't make any sense without some extra invisible stuff.

Since Rubin's time, the evidence for dark matter has only strengthened, partly because we now know how important it was in the early universe, but it has remained stubbornly hard to directly detect, on account of being apparently uninterested in interacting with our particle detectors. The leading idea is that dark matter is some kind of as yet undiscovered fundamental particle that has mass (and therefore gravity) but doesn't have anything to do with electromagnetism or the strong nuclear force. Theories suggest it might be able to interact with other particles via the weak nuclear force, opening up some possibilities for detection, but the signal would be hard to find and we haven't yet seen it. What we *have* seen is a massive amount of evidence for its gravitational impact on stars and galaxies, and on the ability for stars and galaxies to form out of the primordial soup in the first place. And even better than that, we can see evidence for dark matter's existence in the shape of space itself.

One of Einstein's big insights (among many) was that gravity isn't best understood as a force between objects, but rather as the bending of space around anything that has mass. Imagine rolling a tennis ball across the surface of a trampoline. Now put a bowling ball in the center. The way the tennis ball falls toward the bowling ball, or curves as it goes past it, is a pretty good analogy for how objects move through space in the presence of large masses. The shape of the space itself is causing the object's trajectory to curve. But it's not just the paths of massive objects that are affected by the bending of space—even light responds to the shape of the space it's